

The non-radial Hilbert-matrix norm problem

arXiv automatic math run `fa_banach_001`

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Source and question

The source paper is Bellavita–Stylogiannis, *Hilbert matrix operator Acting between conformally invariant spaces*, arXiv:2408.01060 [1]. Its final section records the following question.

8. OPEN QUESTIONS

We note that for every radial measure μ that satisfies (2.1),

$$\|\mathcal{H}\|_{H^\infty \rightarrow M(\mathcal{D}_\mu)} = 1 + \left(\int_{\mathbb{D}} \frac{4}{|1 - z^2|^2} U_\mu(z) dA(z) \right)^{1/2}.$$

However we are not able to compute the exact value of $\|\mathcal{H}\|_{H^\infty \rightarrow M(\mathcal{D}_\mu)}$ in general.

Question 8.1. What is the exact norm of $\mathcal{H}$ from H^∞ to $M(\mathcal{D}_\mu)$ for non-radial μ ?

For a positive Borel measure μ on $\mathbb{D}$, the source assumes

$$\int_{\mathbb{D}} (1 - |w|^2) d\mu(w) < \infty$$

and sets

$$U_\mu(z) = \int_{\mathbb{D}} \log \left| \frac{1 - \bar{w}z}{z - w} \right|^2 d\mu(w).$$

The norm in $M(\mathcal{D}_\mu)$ is

$$\|g\|_{M(\mathcal{D}_\mu)} = |g(0)| + \sup_{\phi \in \text{Aut}(\mathbb{D})} \left(\int_{\mathbb{D}} |g'(\phi(z))|^2 U_\mu(\phi(z)) dA(z) \right)^{1/2}.$$

Result

Theorem 1. Let μ be a nonzero positive Borel measure on $\mathbb{D}$ satisfying

$$\int_{\mathbb{D}} (1 - |w|^2) d\mu(w) < \infty.$$

Then $\mathcal{H} : H^\infty \rightarrow M(\mathcal{D}_\mu)$ is bounded if and only if $\mu(\mathbb{D}) < \infty$. In the bounded case,

$$\|\mathcal{H}\|_{H^\infty \rightarrow M(\mathcal{D}_\mu)} = 1 + \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})}.$$

Consequently Question 8.1 has the above exact answer for arbitrary non-radial μ in the source class: finite mass gives the displayed norm, while infinite mass gives unboundedness.

Proof intuition

The $M(\mathcal{D}_\mu)$ -seminorm is an average, over w , of usual local BMOA Green seminorms after applying an automorphism. A degenerating automorphism σ_r , $r \uparrow 1$, sends every fixed point $w \in \mathbb{D}$ to the same boundary point. For both $L(z) = \log(1/(1-z))$ and $K(z) = \mathcal{H}(1)(z)$, the local BMOA seminorm at $\sigma_r(w)$ tends to the extremal value $\pi/\sqrt{2}$.

If μ has infinite total mass, apply this boundary degeneration on finite-measure truncations of arbitrarily large mass. The seminorm of $\mathcal{H}(1)$ is forced to be infinite, so $\mathcal{H}$ is not bounded. If μ is finite, dominated convergence gives the exact seminorm $(\pi/\sqrt{2})\sqrt{\mu(\mathbb{D})}$, and the standard derivative identity $(1-z)(\mathcal{H}f)' = b_f$, $\|b_f\|_\infty \leq \|f\|_\infty$, gives the sharp upper bound. Testing $f \equiv 1$ gives equality.

Proof

Write

$$G(z, w) = \log \left| \frac{1 - \bar{w}z}{z - w} \right|^2$$

and define the local Green seminorm

$$\mathcal{D}_a(g) = \int_{\mathbb{D}} |g'(z)|^2 G(z, a) dA(z), \quad a \in \mathbb{D}.$$

Green's function is invariant under automorphisms:

$$G(\phi(z), w) = G(z, \phi^{-1}(w)), \quad \phi \in \text{Aut}(\mathbb{D}).$$

Therefore Tonelli's theorem gives, for every analytic g for which either side is finite,

$$\int_{\mathbb{D}} |g'(z)|^2 U_\mu(\phi(z)) dA(z) = \int_{\mathbb{D}} \mathcal{D}_{\phi^{-1}(w)}(g) d\mu(w). \quad (1)$$

We shall use the two boundary-local facts already implicit in the source paper's proof of its $H^\infty \rightarrow \text{BMOA}$ norm estimate. Let

$$L(z) = \log \frac{1}{1-z}, \quad K(z) = \mathcal{H}(1)(z) = \frac{1}{z} \log \frac{1}{1-z}$$

with $K(0) = 1$. Then

$$\sup_{a \in \mathbb{D}} \mathcal{D}_a(L)^{1/2} = \sup_{a \in \mathbb{D}} \mathcal{D}_a(K)^{1/2} = \frac{\pi}{\sqrt{2}}, \quad (2)$$

and, for every fixed $w \in \mathbb{D}$,

$$\mathcal{D}_{\sigma_r(w)}(L)^{1/2} \rightarrow \frac{\pi}{\sqrt{2}}, \quad \mathcal{D}_{\sigma_r(w)}(K)^{1/2} \rightarrow \frac{\pi}{\sqrt{2}} \quad (3)$$

as $r \uparrow 1$, where $\sigma_r(z) = (r - z)/(1 - rz)$. For K , this is the source's decomposition

$$zK'(z) = -\frac{1}{z} \log \frac{1}{1-z} + \frac{1}{1-z};$$

the logarithmic term is negligible in the degenerating local Green seminorm, and the second term has the same boundary limit as L' .

First suppose that $\mu(\mathbb{D}) = \infty$. The standing hypothesis implies that μ is sigma-finite: the closed disks $\{|w| \leq 1 - 1/n\}$ have finite μ -measure and exhaust $\mathbb{D}$. Hence, for every $M > 0$, there is a measurable set $E \subset \mathbb{D}$ with $0 < \mu(E) < \infty$ and $\mu(E) > M$. Using (1), (3), and the uniform bound in (2), dominated convergence gives

$$\lim_{r \uparrow 1} \int_E \mathcal{D}_{\sigma_r(w)}(K) d\mu(w) = \frac{\pi^2}{2} \mu(E).$$

Therefore the $M(\mathcal{D}_\mu)$ -seminorm of $K = \mathcal{H}(1)$ is at least $(\pi/\sqrt{2})\sqrt{\mu(E)}$. Since M was arbitrary, this seminorm is infinite. Thus $\mathcal{H}(1) \notin M(\mathcal{D}_\mu)$, and $\mathcal{H}$ is not bounded from H^∞ into $M(\mathcal{D}_\mu)$.

Now suppose that $\mu(\mathbb{D}) < \infty$. Applying (1)–(3) directly over all of $\mathbb{D}$, again by dominated convergence, gives

$$\sup_{\phi \in \text{Aut}(\mathbb{D})} \left(\int_{\mathbb{D}} |L'(z)|^2 U_\mu(\phi(z)) dA(z) \right)^{1/2} = \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})}. \quad (4)$$

The same identity holds with L replaced by K .

For $f \in H^\infty$, Bellavita–Stylogiannis prove

$$(1 - z)(\mathcal{H}f)'(z) = b_f(z), \quad b_f \in H^\infty, \quad \|b_f\|_\infty \leq \|f\|_\infty.$$

Also,

$$\mathcal{H}f(0) = \int_0^1 f(t) dt.$$

Using (4),

$$\|\mathcal{H}f\|_{M(\mathcal{D}_\mu)} \leq \left(1 + \frac{\pi}{\sqrt{2}} \sqrt{\mu(\mathbb{D})} \right) \|f\|_\infty.$$

Thus the displayed constant is an upper bound for the operator norm. Taking $f \equiv 1$ gives $\mathcal{H}f = K$, $K(0) = 1$, and the K -version of (4), so the same constant is attained. This proves the formula and the theorem.

Verification notes and limitations

The proof is for the U_μ -based $M(\mathcal{D}_\mu)$ norm displayed in the source paper. The most important reviewer checks are:

- the automorphism-invariance identity for the Green kernel;
- the boundary-local limits (3), especially for $K = \mathcal{H}(1)$;
- the sigma-finite truncation argument in the infinite-mass case.

This packet supersedes the earlier finite-measure partial packet. The theorem also makes the finite-mass hypothesis unavoidable for boundedness in this norm convention.

The local run indexes had only the earlier finite-measure partial packet for arXiv:2408.01060. A bounded web search on 2026-06-18 for the arXiv id, the paper title, “Question 8.1”, “non-radial”, “Hilbert matrix”, and $M(\mathcal{D}_\mu)$ found the source paper and nearby Hilbert-matrix papers/surveys, but no later exact answer to Question 8.1.

Human-review recommendation

Review as a likely valid full answer to Question 8.1 under the source’s displayed $M(\mathcal{D}_\mu)$ norm. The subtle point is compatibility with the source’s later examples/norm conventions for Q_p -type spaces; the packet’s proof uses the U_μ -Green-potential norm exactly as displayed in the question’s section.

References

- [1] C. Bellavita and G. Stylogiannis, *Hilbert matrix operator Acting between conformally invariant spaces*, arXiv:2408.01060, 2024.